Akoizo Original Practice - Form 5

Division B | Hovercraft | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

Name: _____ Date: _____

1. (2 points) How may rolls of nickels be attached to the vehicle?

Unit: Carrying nickels: mass, balance, and security.
Worked-case data: A vehicle carries 3 rolls, the half roll, and 12 loose nickels, but 2 loose nickels fall out mid-run. The run is complete. Find the Mass Score and the penalty effect.
Separate investigation observations:
Run 1: 4 rolls in cradles, 20 loose in an open tray: 3 nickels lost at start.
Run 2: Same load, tray with 2 cm walls: nothing lost.
Run 1 score effect: Penalty x 0.8 applied to the whole run.

A. Held in a shaped cradle without tape or adhesive

B. Taped to the deck

C. Unrolled into a tray

D. Glued in place

Answer: _____

2. (2 points) Why can starting farther behind the start line shorten the timed run?

A. The vehicle builds speed before timing begins

B. The track is shorter

C. The supervisor starts timing later

D. It reduces vehicle mass

Answer: _____

3. (2 points) A team uses the ramp and finishes exactly on TT carrying 2 rolls. What is the run score?

A. 60

B. 100

C. 80

D. 48

Answer: _____

4. (2 points) Why are papers with calibration data impounded with the vehicle?

Unit: Hitting the Target Time: calibration and repeatability.
Worked-case data: Your curve shows setting 3 -> 9.6 s and setting 4 -> 11.2 s (full load, 5.0 V). The Target Time is 10.0 s. Which setting do you choose?
Separate investigation observations:
Calibration: Settings 2, 4, 6 gave 8.0, 12.0, 16.0 s.
Target Time: 13.0 s announced.
Run 1 at 4.5: 13.5 s.

A. They replace the run

B. They are a tiebreaker

C. No additional notes may be brought in after impound

D. The supervisor grades them

Answer: _____

5. (4 points) Why is a microcontroller timer not an option in this event?

Unit: The electrical system: batteries, motors, and speed control.
Worked-case data: A team wants to use eight 1.5 V alkaline AA cells. Is that legal, and what are the options?
Separate investigation observations:
Monday: Fresh pack 5.2 V, knob at 3: 11.8 s.
Wednesday: Same knob, pack at 4.7 V: 13.4 s.
Wednesday, recharged: Pack 5.2 V, knob at 3: 11.9 s.

Answer: _____
_____
_____

6. (4 points) Why does a larger footprint reduce the fan power needed?

Unit: Levitation: the air cushion and the skirt.
Worked-case data: A hovercraft with nickels weighs 1.6 kg and has a 0.35 m x 0.35 m cushion. What cushion pressure is needed?
Separate investigation observations:
Empty test: Vehicle glides freely across the floor with lift on.
With all nickels: Rear skirt drags and the craft stops quickly.
Nickels moved forward 6 cm: Glides freely again, sitting level.

Answer: _____
_____
_____

7. (4 points) What should a team do if inspection finds the dowel 4 cm from the front?

Answer: _____
_____
_____

8. (4 points) Explain how an off-center load can slow the vehicle.

Unit: Carrying nickels: mass, balance, and security.
Worked-case data: A vehicle carries 3 rolls, the half roll, and 12 loose nickels, but 2 loose nickels fall out mid-run. The run is complete. Find the Mass Score and the penalty effect.
Separate investigation observations:
Run 1: 4 rolls in cradles, 20 loose in an open tray: 3 nickels lost at start.
Run 2: Same load, tray with 2 cm walls: nothing lost.
Run 1 score effect: Penalty x 0.8 applied to the whole run.

Answer: _____
_____
_____

9. (4 points) Why is a vehicle that barely reaches the finish risky?

Unit: Thrust, friction, and motion down the track.
Worked-case data: A hovercraft must cross 185 cm in 12.0 s at steady speed. It enters the timed zone already at speed. What speed is required, and how does adding 0.2 kg affect it?
Separate investigation observations:
Trial 1: Start 5 cm behind line: 14.1 s.
Trial 2: Start 40 cm behind line: 11.6 s.
Trial 3: Start 40 cm behind, plus 2 rolls: 13.8 s.

Answer: _____
_____
_____

10. (6 points) Solve this case and explain your reasoning: TT is 12.0 s. A vehicle carries 3 rolls and 10 loose nickels and finishes in 13.2 s with nothing falling off. Find the run score.

Answer: _____
_____
_____

11. (6 points) Solve this case and explain your reasoning: Your curve shows setting 3 -> 9.6 s and setting 4 -> 11.2 s (full load, 5.0 V). The Target Time is 10.0 s. Which setting do you choose?

Answer: _____
_____
_____

12. (2 points) Why did Wednesday's first run take longer?

Supplied synthetic observations:
Monday: Fresh pack 5.2 V, knob at 3: 11.8 s.
Wednesday: Same knob, pack at 4.7 V: 13.4 s.
Wednesday, recharged: Pack 5.2 V, knob at 3: 11.9 s.

A. The pack voltage was lower

B. The knob drifted to a new setting

C. The track was longer

Answer: _____

13. (2 points) Define "Cushion pressure" in the context of Hovercraft.

Answer: _____
_____
_____

14. (2 points) Define "Impound" in the context of Hovercraft.

Answer: _____
_____
_____

15. (2 points) Define "Half roll" in the context of Hovercraft.

Answer: _____
_____
_____

16. (2 points) Define "Acceleration" in the context of Hovercraft.

Answer: _____
_____
_____

17. (2 points) Define "Mass Score (MS)" in the context of Hovercraft.

Answer: _____
_____
_____

18. (2 points) Define "Accuracy" in the context of Hovercraft.

Answer: _____
_____
_____